11.[**Open Ended- SCAN Algorithm**]

Consider a disk queue that contains requests for input/output operations at cylinder numbers **98, 183, 41, 122, 14, 124, 65, and 67**.The disk head is initially positioned at **cylinder 53**, and the cylinders are numbered from **0 to 199**.Use the **SCAN (Elevator) disk scheduling algorithm** to service these requests by moving the disk head in a particular direction and then reversing its direction after reaching the end.Write a program to determine the **total head movement**, measured in the number of cylinders traversed, while processing all the given requests

CODE :

```
 GNU nano 8.7                          scan.c                          Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total = 0;

    // Sorting
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(req[i] > req[j]){
                int temp = req[i];
                req[i] = req[j];
                req[j] = temp;
            }
        }
    }

    int pos = 0;
    for(int i=0;i<n;i++){
        if(head < req[i]){
```

```
GNU nano 8.7                    scan.c                    Modified

    int pos = 0;
    for(int i=0;i<n;i++){
        if(head < req[i]){
            pos = i;
            break;
        }
    }

    printf("Order of servicing:\n");

    // Move RIGHT
    for(int i=pos;i<n;i++){
        printf("%d -> ", req[i]);
        total += abs(head - req[i]);
        head = req[i];
    }

    // Go to end
    total += abs(head - 199);
    head = 199;

    // Move LEFT
```

```
GNU nano 8.7                    scan.c                    Modified
    // Move RIGHT
    for(int i=pos;i<n;i++){
        printf("%d -> ", req[i]);
        total += abs(head - req[i]);
        head = req[i];
    }

    // Go to end
    total += abs(head - 199);
    head = 199;

    // Move LEFT
    for(int i=pos-1;i>=0;i--){
        printf("%d -> ", req[i]);
        total += abs(head - req[i]);
        head = req[i];
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

OUTPUT :

```
M ~

Sakshi Wadiyalwar@DESKTOP-JJN908K MINGW64 ~
$ nano scan.c

Sakshi Wadiyalwar@DESKTOP-JJN908K MINGW64 ~
$ gcc scan.c -o scan

Sakshi Wadiyalwar@DESKTOP-JJN908K MINGW64 ~
$ ./scan
Total Head Movement = 236

Sakshi Wadiyalwar@DESKTOP-JJN908K MINGW64 ~
$
```